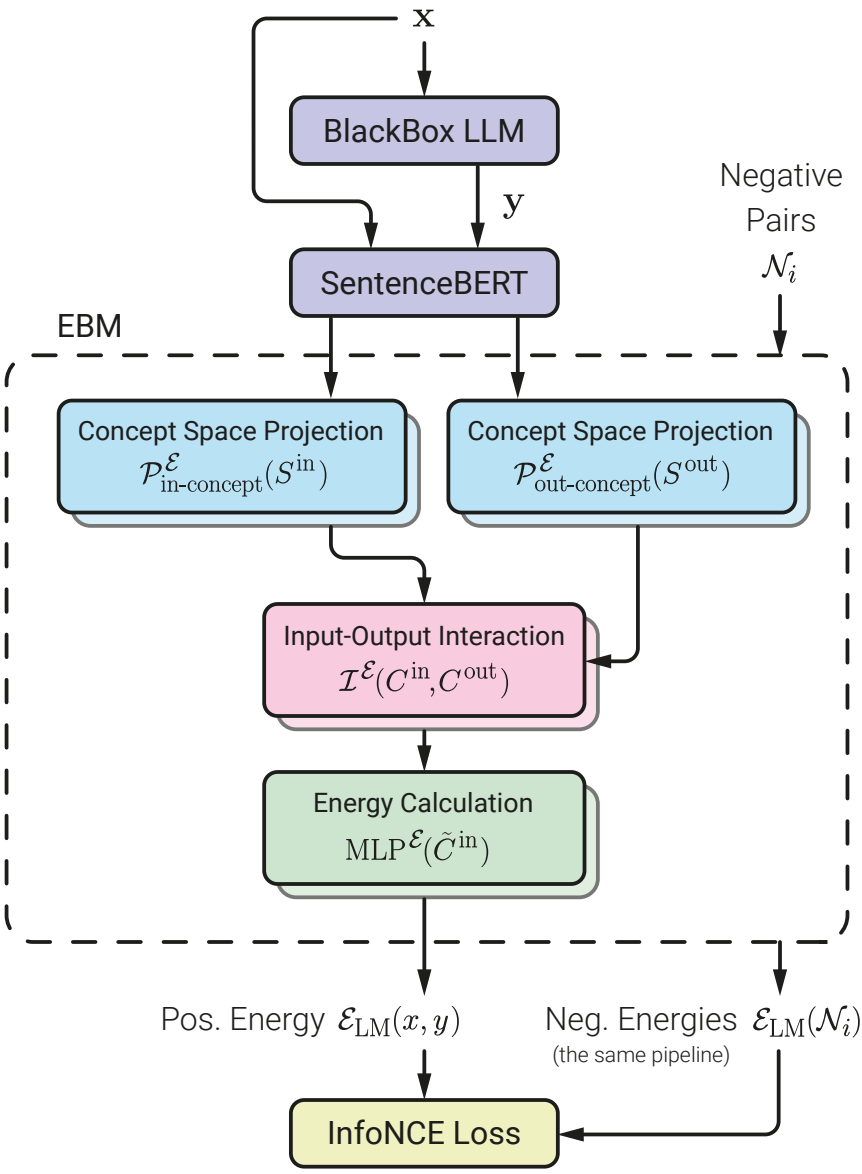

x
BlackBox LLM
y
Negative Pairs
$\mathcal{N}_i$
SentenceBERT
EBM
Concept Space Projection
$\mathcal{P}^{\mathcal{E}}_{\text{in-concept}}(S^{\text{in}})$
Concept Space Projection
$\mathcal{P}^{\mathcal{E}}_{\text{out-concept}}(S^{\text{out}})$
Input-Output Interaction
$\mathcal{I}^{\mathcal{E}}(C^{\text{in}}, C^{\text{out}})$
Energy Calculation
$\text{MLP}^{\mathcal{E}}(\tilde{C}^{\text{in}})$
Pos. Energy $\mathcal{E}_{\text{LM}}(x, y)$
Neg. Energies $\mathcal{E}_{\text{LM}}(\mathcal{N}_i)$
(the same pipeline)
InfoNCE Loss